Virtual Ink Escrow Engine Build Specification

A provider-neutral settlement engine for a web-first marketplace

Document detail	Value
Prepared for	Shadow Root Security Technologies
Product	Virtual Ink - Your Ideas. Our Print. Delivered.
Current delivery model	Web-first, API-centred, mobile-ready
First outcome	A tested ledger and order-state engine that can later connect to approved payment providers
Scope status	Build scope - simulation and manual verification before live money movement
Revision	8 October 2026

How to use this document

This document is a build and review baseline. It separates confirmed foundation work from later marketplace, payment, delivery, and mobile work. Every sensitive feature must pass its acceptance gate before it is promoted.

1. Escrow purpose and boundary

The Virtual Ink escrow engine coordinates customer payment status, vendor production, delivery proof, release, refund, and dispute records. It should not be treated as permission to hold regulated client funds until the payment structure, provider, terms, and local compliance position are approved.

- Build the ledger and state machine first using test transactions or manual verification.
- Keep the payment provider behind an adapter so the domain layer remains portable.
- Never store card numbers, CVV values, provider secrets, or unnecessary financial data.
- Use integer minor units and an explicit currency code; never floating-point money arithmetic.
- Every release, refund, dispute, and operator override requires an audit event.

2. Escrow domain architecture

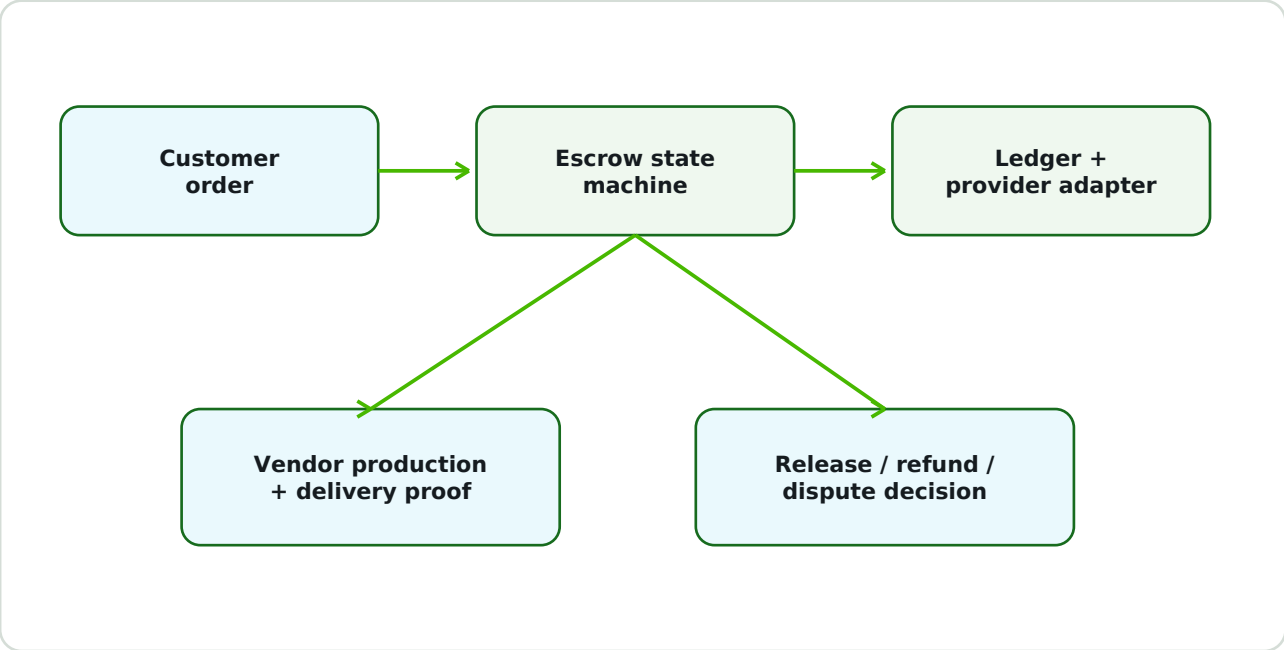


Figure 1. The escrow engine is part of the order domain. The ledger records money facts; the state machine records whether production, handover, release, refund, or dispute is permitted.

3. Core records

Record	Required fields	Owner
Escrow account	escrow_id, order_id, currency, status, version	Virtual Ink
Ledger entry	entry_id, escrow_id, type, amount_minor, currency, reference, created_at	Virtual Ink
Payment attempt	attempt_id, provider_ref, idempotency_key, status, verified_at	Provider adapter
Release instruction	release_id, trigger, recipient, amount, approved_by, status	Operator or policy
Dispute	dispute_id, reason, evidence_refs, status, decision, decided_by	Operator
Reconciliation record	period, provider totals, internal totals, difference, resolution	Operator

4. Escrow state machine

State	Allowed next states	Meaning
AWAITING_PAYMENT	PAYMENT_PENDING, CANCELLED	Quote accepted; payment has not been verified.
PAYMENT_PENDING	FUNDS_HELD, FAILED, CANCELLED	Provider or operator is verifying payment.
FUNDS_HELD	IN_PRODUCTION, REFUND_PENDING, DISPUTED	Payment is recorded as held for the order.
IN_PRODUCTION	READY_FOR_HANDOVER, DISPUTED, CANCELLED	Vendor is producing the work.
READY_FOR_HANDOVER	DELIVERED_OR_COLLECTED, DISPUTED	Work is ready and handover is pending.
DELIVERED_OR_COLLECTED	RELEASE_PENDING, DISPUTED	Handover evidence is recorded.
RELEASE_PENDING	RELEASED, DISPUTED	Release is awaiting approved rule or operator action.
RELEASED	Terminal	Vendor settlement has been recorded.
REFUND_PENDING	REFUNDED, DISPUTED	Refund is awaiting verification or provider action.

5. Order-to-release flow

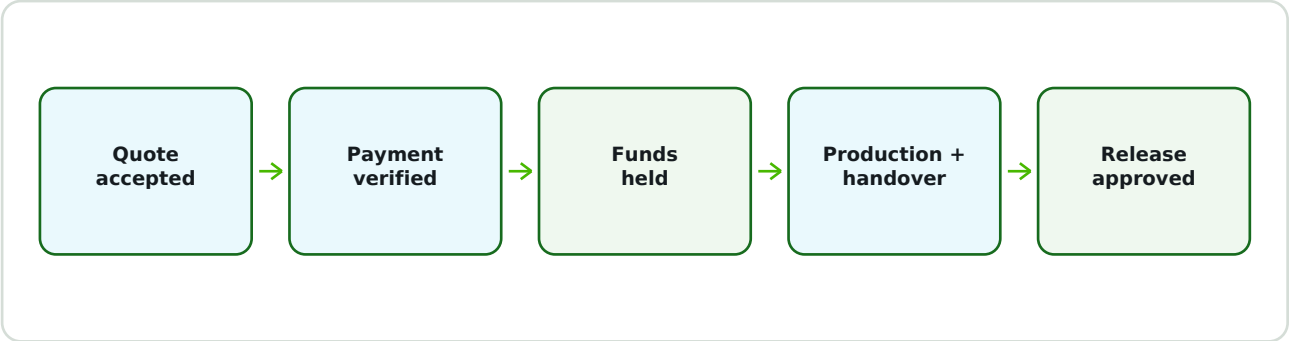


Figure 2. Release must follow a verified business event. The first implementation may use a manual approval record instead of live provider movement.

Release conditions

- Customer acceptance, delivery proof, or collection confirmation is recorded.
- The order is not in dispute, cancellation, or refund processing.
- The released amount equals the approved settlement calculation.
- Any platform fee, vendor amount, delivery amount, or refund is visible in the ledger.
- An operator override requires a reason, actor, timestamp, and supporting evidence.

6. Ledger rules

Rule	Implementation requirement
Append-only facts	Do not edit a posted ledger entry; create a correcting entry.
Balanced movement	Every hold, release, fee, refund, or adjustment has a traceable counterpart.
Currency safety	Store currency with each amount and reject mismatched currency operations.
Idempotency	A repeated provider callback or operator retry cannot double-release funds.
Versioning	State updates use optimistic concurrency or an equivalent version check.
Reconciliation	Compare provider references and internal entries by period and order.

7. API boundary

Endpoint group	Example behavior	Authorization
Quotes	Accept an approved quote and create payment intent	Customer owns quote
Payment status	Record provider or manual verification	Provider adapter or operator
Escrow status	Read current hold, release, refund, or dispute state	Customer or assigned operator
Release	Create approved release instruction	Operator or approved policy
Refund	Create refund request and track result	Operator; customer request may start review
Dispute	Open, evidence, decide, close	Customer, vendor, or operator by rule
Reconciliation	Compare internal and provider records	Operator only

8. Failure and dispute handling

- Payment callback arrives twice: accept the first idempotently and record the duplicate safely.
- Vendor cannot complete production: keep funds held or begin refund review; do not silently release.
- Customer disputes output: freeze release and record evidence references.
- Delivery fails: record attempt, return or re-delivery decision, and owner.
- Provider and ledger differ: stop automatic release for affected orders and open reconciliation work.

9. Escrow security controls

Risk	Control	Test
Double release	Idempotency key, ledger uniqueness, state version	Repeat callback test
Unauthorized refund	Operator permission, reason, audit event	Wrong-role denial
Amount manipulation	Server-side calculation and immutable snapshot	Client-total tamper test
Provider spoofing	Signed webhook verification and provider reference	Invalid signature test
Dispute bypass	Release blocked while dispute is open	State transition test
Data leakage	Scoped views and redacted logs	Customer/vendor access test

10. Build sequence

Step	Build slice	Exit evidence
1	Escrow schema and append-only ledger	Migration, constraints, ledger tests
2	Order and escrow state machine	Allowed and denied transitions
3	Manual payment verification adapter	Manual verification record and audit
4	Release, refund and dispute workflows	Operator review and denial tests
5	Reconciliation report	Period comparison and difference handling
6	Sandbox provider adapter	Signed callback, replay and failure tests
7	Mobile payment status surfaces	Shared API contract and secure UI states

11. Open decisions before live money movement

- Which payment provider and settlement account will be used?
- Will Virtual Ink act as marketplace facilitator, agent, or another approved structure?
- What event permits release: customer acceptance, delivery proof, time window, or operator approval?
- Who owns refunds, reprints, failed deliveries, chargebacks, and disputes?
- What fees, vendor settlements, taxes, and currency rules are approved?
- What legal, accounting, privacy, and payment-provider review is required before production?

12. Acceptance checklist

- No live funds are moved in the first escrow build slice.
- Ledger entries are append-only and use minor-unit amounts.
- State transitions are server-controlled and tested.
- Duplicate callbacks cannot duplicate money movement.
- Disputes and refunds block release until resolved.

- Customer, vendor, delivery, and operator views expose only permitted fields.
- Reconciliation and operator overrides are auditable.